

GOA PANAJI, India

WMO: 431920

Lat: 15.4922N

Lon: 73.8252E

Elev: 192

StdP: 14.59

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	67.3	68.6	54.4	63.3	82.2	57.1	70.1	81.2	12.4	82.5	11.3	83.9	3.5	90	0.323

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	11.1	93.6	78.5	92.7	78.7	91.9	78.5	82.8	89.7	82.3	89.0	81.7	88.2	6.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.0	162.3	86.4	80.5	159.7	86.1	80.0	156.8	85.5	47.0	90.3	46.3	89.2	45.7	88.6	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	13.7	11.7	64.1	96.8	2.5	1.6	62.3	97.9	60.9	98.8	59.5	99.7	57.6	100.9
			58.4	84.1	1.6	1.1	57.2	85.0	56.3	85.6	55.3	86.3	54.2	87.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.0	79.4	80.2	82.7	85.3	86.2	82.2	80.6	80.6	81.0	82.5	82.4	80.5	(6)
	DBStd	2.92	2.14	2.18	2.14	1.72	2.11	3.11	2.24	1.83	1.71	2.02	2.00	2.20	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD50	11669	911	845	1013	1059	1123	965	950	948	930	1008	971	946	(10)
	CDD65	6194	446	425	548	609	658	515	485	483	480	543	521	481	(11)
	CDH74	65737	3861	3996	6151	7907	8843	5716	4701	4565	4587	5682	5353	4375	(12)
	CDH80	25123	1571	1638	2530	3747	4477	2001	1053	923	1140	2021	2201	1821	(13)
(14) Wind	WSAvg	4.3	3.4	3.7	3.7	4.3	4.9	6.0	7.1	5.6	3.8	3.1	3.1	3.1	(14)
(15) Precipitation	PrecAvg	112.0	0.0	0.0	0.1	0.2	3.5	33.4	36.0	22.2	10.5	5.0	1.1	0.5	(15)
	PrecMax	169.9	0.4	0.1	1.7	1.8	28.7	62.4	60.1	47.0	30.1	15.4	7.5	8.5	(16)
	PrecMin	67.8	0.0	0.0	0.0	0.0	0.0	4.4	11.9	3.9	0.9	0.0	0.0	0.0	(17)
	PrecStd	19.8	0.1	0.0	0.3	0.3	5.4	10.7	11.2	9.4	6.4	4.1	1.8	1.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.2	93.7	93.9	93.9	94.7	93.4	88.2	87.4	89.1	93.4	93.8	93.3	(19)
		MCWB	71.8	71.2	75.4	80.1	81.7	81.7	81.4	80.4	80.5	78.1	75.2	72.3	(20)
	2%	DB	90.8	91.6	91.5	92.6	93.6	91.4	86.9	86.3	87.8	91.5	92.0	91.4	(21)
		MCWB	71.9	72.9	78.0	80.5	81.4	81.5	80.9	80.2	79.9	78.6	75.6	73.2	(22)
	5%	DB	89.2	89.8	90.3	91.8	92.6	89.4	85.8	85.4	86.6	90.0	90.8	89.9	(23)
		MCWB	72.0	73.8	78.3	80.2	81.1	81.3	80.6	79.9	79.5	79.1	75.8	73.0	(24)
	10%	DB	87.6	88.1	89.2	90.9	91.7	87.5	84.6	84.4	85.5	88.4	89.5	88.5	(25)
		MCWB	72.0	74.3	78.2	79.8	80.6	80.9	80.1	79.6	79.2	79.2	75.9	72.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.3	79.9	81.9	83.4	83.7	83.5	82.4	81.8	81.9	82.7	81.1	79.3	(27)
		MCDB	86.2	87.5	89.7	91.6	91.7	89.4	86.6	85.4	86.7	89.0	88.4	87.6	(28)
	2%	WB	76.5	78.5	81.0	82.3	82.8	82.6	81.5	81.0	81.0	81.6	79.9	77.9	(29)
		MCDB	84.6	85.8	88.5	90.2	90.6	88.3	85.5	84.8	85.6	87.8	87.1	85.7	(30)
	5%	WB	75.3	77.3	80.2	81.6	82.3	81.8	80.9	80.4	80.3	80.8	79.0	76.8	(31)
		MCDB	84.0	84.7	87.3	89.3	90.0	87.3	84.7	84.2	84.9	86.7	86.0	84.2	(32)
	10%	WB	74.2	76.2	79.5	80.9	81.6	81.2	80.2	79.8	79.7	80.1	78.3	75.7	(33)
		MCDB	82.9	83.9	86.4	88.4	89.2	86.3	83.8	83.4	84.0	85.6	84.9	83.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	18.8	17.6	14.2	12.1	11.1	8.1	6.5	7.1	8.9	12.2	16.2	18.6	(35)
		MCDBR	20.5	19.4	14.8	12.3	11.7	10.8	7.9	8.4	10.5	14.8	17.6	19.7	(36)
	5% WB	MCWBR	7.6	7.5	6.0	5.0	4.0	4.3	4.2	4.2	4.5	5.5	6.1	7.2	(37)
		MCWBR	17.9	16.0	12.7	11.3	11.1	9.5	7.5	7.8	9.3	12.1	14.2	15.8	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.459	0.472	0.553	0.607	0.582	0.610	0.659	0.601	0.525	0.575	0.515	0.471	(40)
		taud	2.181	2.158	1.969	1.873	1.942	1.892	1.770	1.905	2.120	1.950	2.083	2.160	(41)
	Ebn at Noon	Ebn at Noon	263	266	247	234	236	227	217	232	251	234	244	255	(42)
		Edh at Noon	45	48	60	66	61	63	71	63	51	59	49	45	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1701	1945	2108	2181	2046	1321	1205	1350	1557	1625	1617	1607	(44)
	RadStd	RadStd	62	55	76	65	118	140	180	113	142	94	88	62	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (4 neighbors)	+0.27	N/A	+1.55	+0.67	+0.65	+0.61	N/A	N/A	N/A	N/A

Nomenclature: See separate page